GROUP NO. 4

SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION CONTROL

PROBLEM STATEMENT

Smart Traffic Management System for Urban Congestion

SIH 25050 – Software edition – Government of Odisha

- **Dumb Timers:** Traditional traffic lights run on fixed countdown timers, meaning they don't care if a road is completely empty or heavily jammed.
- **Wasted Time & Fuel:** Cars end up waiting at red lights even when there is no cross-traffic, which causes unnecessary delays and pollution.

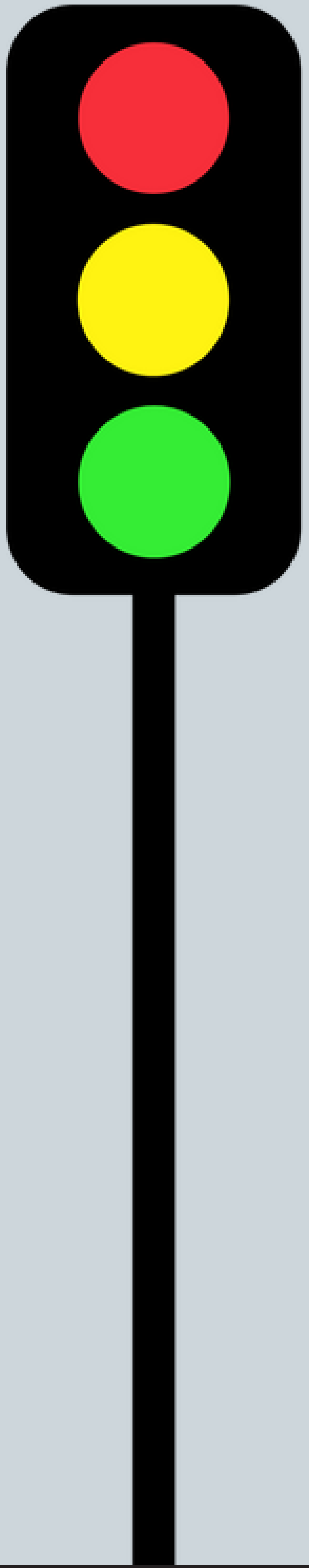
TRADITIONAL APPROCH

- Fixed Timers: Traffic lights run on a set countdown, changing colors even if no one is there.

- Blind System: Because the lights cannot "see" the cars, you often get stuck waiting at a red light while the green light is given to a completely empty road.

- No Emergency Priority: The system cannot recognize approaching ambulances or fire trucks, forcing them to get stuck in normal traffic or dangerously push their way through red lights.

HOW WE ARE GOING TO SOLVE IT



LIVE VEHICLE COUNTING:

We use smart cameras and AI to continuously count exactly how many cars are waiting in each lane at the intersection in real-time.

SMART GREEN LIGHTS

If our system counts 20 cars waiting in the North lane and 0 in the East lane, it immediately gives the green light to the busy lane and skips the empty one, eliminating wasted time.

VEHICLE TYPE RECOGNITION

While counting, the system also analyzes the shapes of the vehicles to identify exactly what is on the road, specifically looking out for emergency vehicles like ambulances or fire trucks.

AUTO EMERGENCY OVERRIDE

The moment an ambulance is recognized, the system ignores the normal counting rules and instantly turns that specific lane green, clearing a safe and fast path for the emergency vehicle.

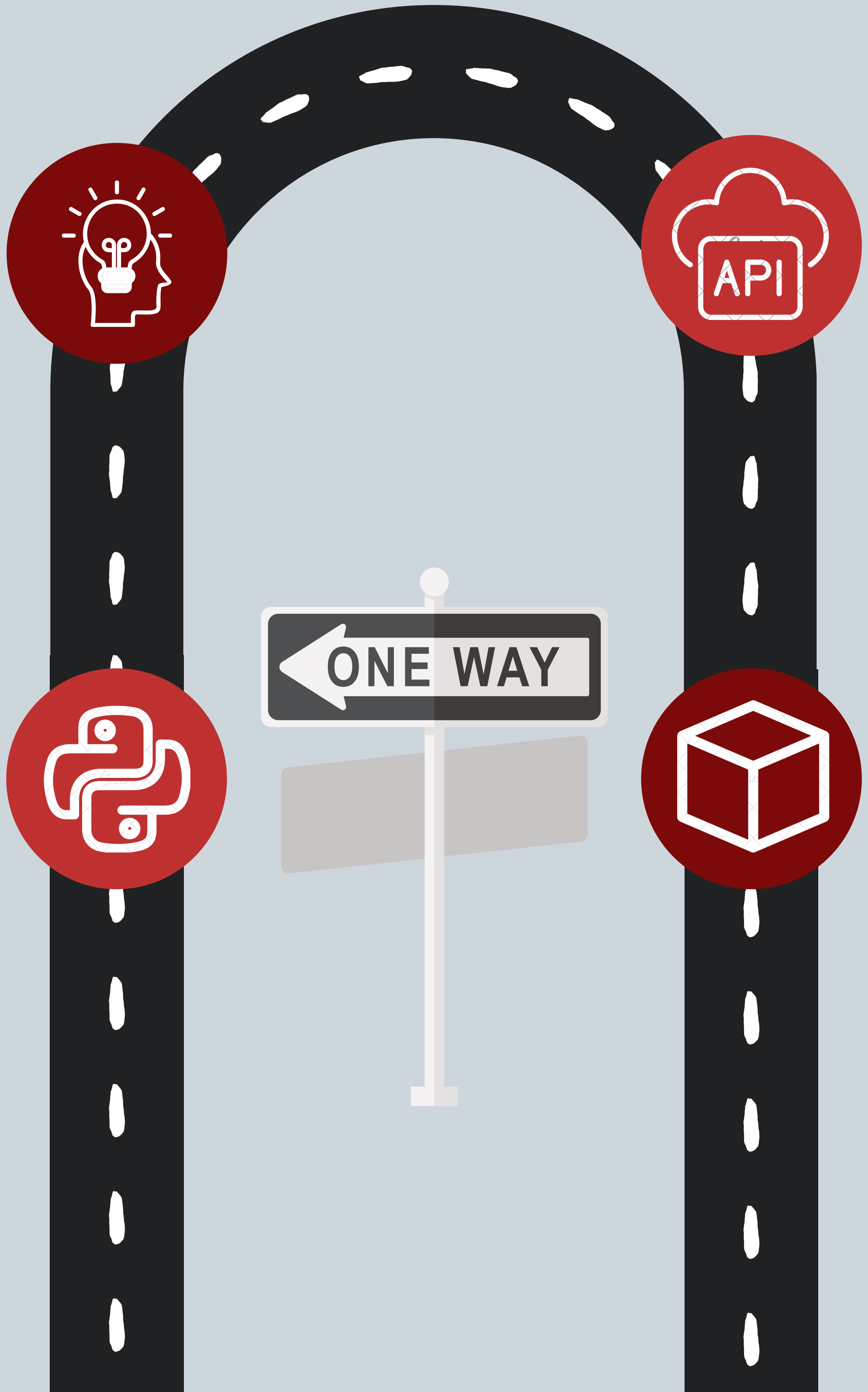
TECHNOLOGY USED

SUMO Simulator:

The traffic engine used to draw the road network. It handles all the background math and acts as our virtual "sensor," perfectly counting the vehicles and identifying which one is an ambulance.

Python (VS Code):

The "brain" of the project. Here, we write our custom Rule-Based AI (the "If-Then" logic) to analyze the counts and vehicle types coming from SUMO.



TraCI API:

The communication bridge. It is the Python library that extracts the vehicle counts from SUMO and sends back our commands to instantly change the virtual traffic lights.

Unity 3D:

The graphics engine. It takes the working traffic logic from SUMO and turns it into a high-quality, realistic 3D visual world so we can clearly see the cars and ambulances moving.

SUMO

WORKFLOW

Step 1:

Use SUMO's Netedit tool to draw the basic roads, lanes, and intersections.

Step 2:

Write the Python code to check how many cars are waiting and change the lights accordingly.

Step 3:

Run the system in SUMO's basic 2D viewer first to make sure the logic actually works.

Step 4:

Send the data to Unity to view the exact same traffic simulation in realistic 3D.

RESEARCH WORK

Name : AI Based Traffic Management System

Publisher : IJEPE National College of Engineering, Tribhuvan University

Key Objectives : System Development , Dynamic Signal Adjustment, Traffic Optimization , Real Time Responsiveness

Our contribution : Co-Simulation , Custom AI Logic , Emergency Priority , Safe 3D Testing

FUTURE SCOPE

Real-World Testing

Replace the virtual SUMO data with actual street cameras and physical traffic lights.

Machine Learning Upgrade

Upgrade your custom "If-Then" rules to a self-learning AI that gets smarter over time.

City-Wide Expansion

Scale the project from a single intersection to a fully synchronized, city-wide traffic network.

Smart Vehicle Communication

Add features where modern cars can "talk" directly to the traffic lights to adjust their speed and avoid stopping.

CONCLUSION

- **Smart Automation:** Replaced static timers with AI that reacts to live traffic.
- **Less Delay:** Reduced waiting times and prioritized emergency vehicles.
- **3D Simulation:** Used SUMO and Unity to safely test rules in a realistic virtual environment.
- **Future Ready:** Proved the concept is ready for real-world smart city deployment.


THANK
YOU